

A. Power

How can you supply energy ?



Energy = Power x Time

$$E = Pt$$

Joule (J) = Watt (W) Second (s)

Problem 1

A heater has a power rating of 2 000 W. What is the maximum amount of energy given out in 1 hour?

$$\begin{aligned} E &= Pt = (2000)(1 \times 60 \times 60) \\ &= 7\,200\,000 \text{ J} \end{aligned}$$

B. Specific Heat Capacity



1kg

HEAT NEEDED



1°C

Specific Heat Capacity =
amount of **heat energy** needed
to raise the temperature
of **1 kg** of a material by **1°C**

$$E = mc\Delta T$$

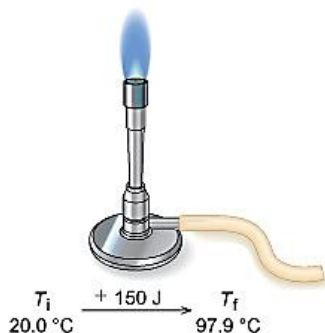
C (Specific Heat Capacity)

$$\begin{aligned} &= \frac{\text{Energy (E)}}{\text{Mass (m)} \times \text{Temp}(\Delta T)} \\ &= \text{J kg}^{-1}\text{°C}^{-1} \end{aligned}$$

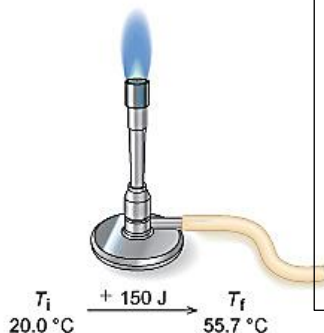
Temp. change (ΔT) = Final – Initial Temp

Problem 2

5 g copper



5 g glass

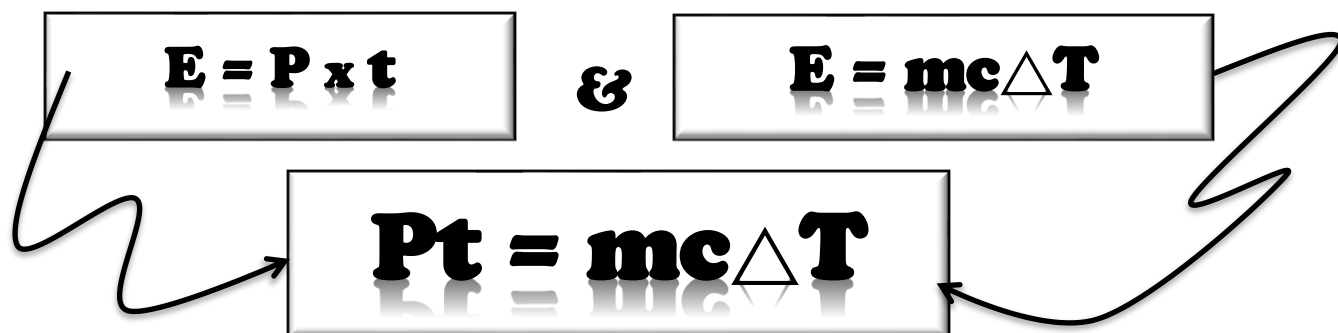


Conclusion



The temperature change of the copper sample is much greater than that of the glass sample. This indicates that copper has

a smaller specific heat capacity.



3. What is the heat required to raise the temperature of 20 kg of a metal from 100°C to 600°C?

(The specific heat capacity of the metal is 100J kg⁻¹ °C⁻¹)

$$E = mc\Delta T = (20)(100)(600 - 100) = 1\,000\,000 \text{ J}$$

4. If 900 J of energy is needed to raise 0.1kg of a metal from 10°C to 20°C. What is the specific heat capacity of the metal?

$$E = mc\Delta T \\ 900 = (0.1)(c)(20 - 10) \\ c = 900 \text{ J kg}^{-1} \text{ °C}^{-1}$$

5. An electric heater in a kettle is used to heat up 0.5 kg of water. It takes 10 minutes to raise the water temperature from 25°C to 60°C. Find the power of the heater.
(Specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹)

$$Pt = mc\Delta T \\ P(10 \times 60) = (0.5)(4200)(60 - 25) \\ P = 122.5 \text{ W}$$

6. A 100 W heater is used to heat a 0.5 kg metal for 10 s. If the initial temperature of the metal is 20°C, Find the final temperature of the metal.
(Specific heat capacity of metal = 200J kg⁻¹ °C⁻¹)

$$Pt = mc\Delta T \\ (100)(10) = (0.5)(200)(T_f - 20) \\ T_f = 30 \text{ °C}$$

7. An immersion heater is used to heat a 0.5 kg steel for 1 minutes. If the temperature raise from 20°C to 200°C, Find the power of the heater. (Specific heat capacity of steel = 480 J kg⁻¹ °C⁻¹)

$$Pt = mc\Delta T \\ P(1 \times 60) = (0.5)(480)(200 - 20) \\ P = 720 \text{ W}$$

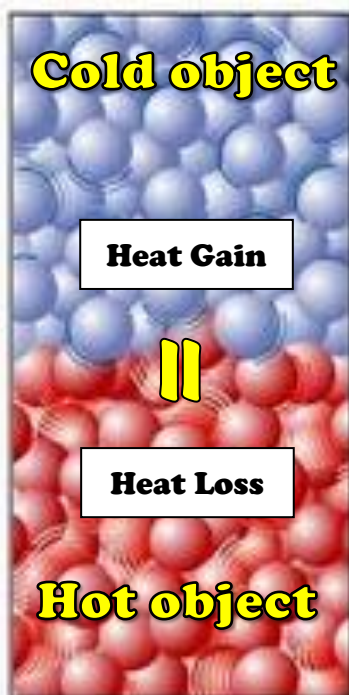
8. A hot water system is used to heat up 20 kg of water from 60 °C to 90 °C. Find
(a) the energy transferred to the water; and
(b) the time required if energy is transferred to the water at a rate of 1200 W

- (a) Applying $E = mc\Delta T$, we have
 $E = 20 \times 4200 \times (90 - 60)$
 $= 2.52 \times 10^6 \text{ J}$
The energy transferred to the water is $2.52 \times 10^6 \text{ J}$.

- (b) Applying $P = \frac{E}{t}$, we have
 $t = \frac{2.52 \times 10^6}{1200}$
 $= 2100 \text{ s}$

The time required to heat up the water is 2100 s.

C. Energy Transfer



10°C Water

50°C Water

? °C Water

The energy transferred =

temperature difference between two bodies

Law of conservation of energy

Energy can neither be created nor destroyed
but only changes from one form to the others.

Heat loss
from hot object = **Heat gain**
by cold water

9. What is the final temperature if mix a 200 kg, 80°C water with 100 kg, 20°C cold water?

(Specific heat capacity of water = 4 200 J kg⁻¹ °C⁻¹)

$$\begin{aligned} mc\Delta T (\text{hot water}) &= mc\Delta T (\text{cold water}) \\ (200)(c)(80 - T_f) &= (100)(c)(T_f - 20) \\ 2(80 - T_f) &= (T_f - 20) \\ 160 - 2T_f &= T_f - 20 \\ 3T_f &= 180 \\ T_f &= 60^\circ\text{C} \end{aligned}$$



10. 0.15 kg of cold water at 10°C was mixed with 0.1 kg of hot water. The resulting temperature was 40°C. Find the initial temperature of the hot water. (Specific heat capacity of water = 4 200 J kg⁻¹ °C⁻¹)

$$\begin{aligned} mc\Delta T (\text{hot water}) &= mc\Delta T (\text{cold water}) \\ (0.1)(c)(T - 40) &= (0.15)(c)(40 - 10) \\ 0.1 T - 4 &= (0.15)(30) \\ T &= 85^\circ\text{C} \end{aligned}$$

11. A student pours 0.8 kg of water at 95 °C into a cup noodle. The mass and initial temperature of the noodles are 0.3 kg and 20 °C respectively. Find the temperature of the mixture of the water and noodles. State any assumption you have made.

(The specific heat capacities of water and noodle are 4200 and 2400 J kg⁻¹ °C⁻¹ respectively.)

Assume there is no energy lost to the surroundings. By the law conservation of energy, we have

energy lost by the hot water = energy gained by the noodles

$$3360(95 - T) = 720(T - 20)$$

$$T \approx 81.8^\circ\text{C}$$

12. 0.2 kg of liquid X at 60°C is mixed with water at 20°C. The final temperature is found to be 36°C. If the specific heat capacity of liquid X is 2100 J kg⁻¹ °C⁻¹, find the mass of water mixed with liquid X.

(Specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹)

$$mc\Delta T \text{ (hot Liquid X)} = mc\Delta T \text{ (cold water)}$$

$$(0.2)(2100)(60 - 36) = (m)(4200)(36 - 20)$$

$$m = 0.15 \text{ kg}$$

13. 200 g of hot water at 80 °C is contained in a glass, A, of mass 150 g. Another glass, B, of the same mass contains 200 g of cold water at 20 °C. The specific heat capacity of glass is 670 J kg⁻¹ °C⁻¹.

(Specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹)

- (a) The hot water in glass A is poured into glass B and the mixture is stirred well. What is the final temperature?
(b) The cold water in glass B is poured into glass A and the mixture is stirred well. What is the final temperature?
(c) What should the final temperature be without the effect of the container?
(d) What is your conclusion from the calculation?

$$(a) \quad mc\Delta T \text{ (hot Water)} = mc\Delta T \text{ (cold water)} + mc\Delta T \text{ (cold glass)}$$

$$(0.2)(4200)(80 - T_f) = (0.2)(4200)(T_f - 20) + (0.15)(670)(T_f - 20)$$

$$840(80 - T_f) = (T_f - 20)(840 + 100.5)$$

$$840(80 - T_f) = (T_f - 20)(940.5)$$

$$(80 - T_f) = (T_f - 20)(1.12)$$

$$(80 - T_f) = 1.12T_f - 22.4$$

$$102.4 = 2.12T_f$$

$$T_f = 48.3^\circ\text{C}$$

(c)

$$mc\Delta T \text{ (hot Water)} = mc\Delta T \text{ (cold water)}$$

$$(0.2)(4200)(80 - T_f) = (0.2)(4200)(T_f - 20)$$

$$T_f = 50^\circ\text{C}$$

(b)

$$mc\Delta T \text{ (hot Water)} + mc\Delta T \text{ (hot glass)} = mc\Delta T \text{ (cold water)}$$

$$(0.2)(4200)(80 - T_f) + (0.15)(670)(80 - T_f) = (0.2)(4200)(T_f - 20)$$

$$940.5(80 - T_f) = (T_f - 20)(840)$$

$$1.12(80 - T_f) = (T_f - 20)$$

$$89.6 - 1.12T_f = T_f - 20$$

$$109.6 = 2.12T_f$$

$$T_f = 51.7^\circ\text{C}$$

- (d) The container will affect the final temperature of the mixture.